

Solving a System of Linear Equations - Part 2

Gauss Elimination with Pivoting

Possible problems with naive Gaussian Elimination

Case 1: When the pivot element is zero during the process, it causes the error of 'DIV by 0'. Also, the pivot element very close to zero also causes a large error.

- Example:

$$\begin{array}{l} 0x_1 + x_2 = 1 \\ x_1 + x_2 = 2 \end{array} \rightarrow \text{divide by zero!!}$$

This problem can be solved by the process called Pivoting:

- Exchanging the pivot row with another that has a nonzero pivot element.

Q. Which row equation should be chosen for the pivoting?

The element that has non-zero coefficient

Case 2: When the pivot element is small relative to other terms in the equation

- Example:

$$\begin{array}{l} \varepsilon x_1 + x_2 = 1 \\ x_1 + x_2 = 2 \end{array}$$

if $\varepsilon \ll 1$ or $\varepsilon \approx 0$

then $x_2 \approx 1, x_1 \approx 1$ (true solution)

using Gauss elimination

$$\begin{cases} \varepsilon x_1 + x_2 = 1 \\ 0 + (1 - \frac{1}{\varepsilon})x_2 = 2 - \frac{1}{\varepsilon} \end{cases} \rightarrow \begin{cases} x_2 = \frac{2 - \varepsilon^{-1}}{1 - \varepsilon^{-1}} \approx 1, \\ x_1 = \varepsilon^{-1}(1 - x_2) \approx 0 \text{ (error)} \end{cases}$$

- Example2

$$\begin{cases} x_1 + \varepsilon^{-1}x_2 = \varepsilon^{-1} \\ x_1 + x_2 = 2 \end{cases} \rightarrow \begin{cases} x_1 + \varepsilon^{-1}x_2 = \varepsilon^{-1} \\ (1 - \varepsilon^{-1})x_2 = 2 - \varepsilon^{-1} \end{cases}$$

$$x_2 = \frac{2 - \varepsilon^{-1}}{1 - \varepsilon^{-1}} \approx 1, \quad x_1 = \varepsilon^{-1} - \varepsilon^{-1}x_2 \approx 0 \text{ (error!!)}$$

$\Rightarrow$ Interchange the pivot equations

$$\begin{cases} x_1 + x_2 = 2 \\ x_1 + \varepsilon^{-1}x_2 = \varepsilon^{-1} \end{cases} \Rightarrow \begin{cases} x_1 + x_2 = 2 \\ 0 + (\varepsilon^{-1} - 1)x_2 = \varepsilon^{-1} - 2 \end{cases} \Rightarrow \begin{cases} x_1 \approx 2 - x_2 \approx 1 \\ x_2 = \frac{\varepsilon^{-1} - 2}{\varepsilon^{-1} - 1} \approx 1 \end{cases}$$

These problems (1) Pivot element is zero/close to zero and (2) Pivot element relative small in the equation, can be avoided by Pivoting.

Method 1: Partial pivoting

Find the largest element in the column and use the equation with that largest element as the pivot equation.

- Example:

$$\begin{array}{l} \textcircled{1} \rightarrow \\ \textcircled{2} \rightarrow \end{array} \left[\begin{array}{cc|c} 2 & 2c & 2c \\ 1 & 1 & 2 \end{array} \right]$$

$\rightarrow$ Using $\textcircled{1}$ as the pivot

- But, if c is a large number (e.g. $c = 10,000$) it can cause a problem. Then, we can use the following method.

Method 2: Scaled partial pivoting

Select the (i^{th}) equation with the larger Scaled Partial pivot for the k^{th} pivot.

$$S.P = \frac{|a_{ik}|}{\max_{1 \leq j \leq n} |a_{ij}|} \quad \begin{array}{l} \text{Pivot candidate at } k^{\text{th}} \text{ column} \\ \text{Largest number in } i^{\text{th}} \text{ equation} \end{array}$$

- Example

$$\begin{bmatrix} 0.001 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \begin{array}{l} \max |a_{1j}| = 1 \\ \max |a_{2j}| = 1 \end{array}$$

For $k=1$, S.P of Eq₁=0.001/1, S.P of Eq₂=1/1.

Thus, select Eq2 for the first pivot equation by exchanging eq1 and eq2. This can be done by multiplying a permutation matrix **P**

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2C \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2C \end{bmatrix}$$

Gauss - Jordan elimination

Gauss elimination method was making the equivalent triangular matrix. This can be modified to make the matrix as the reduced row-echelon form of diagonal matrix with ones in the pivot elements

- Example:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & d_1 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 1 & d_3 \end{array} \right)$$

$$\left[\begin{array}{cccc|c} 4 & -2 & -3 & 6 & 12 \\ -6 & 7 & 6.5 & -6 & -6.5 \\ 1 & 7.5 & 6.25 & 5.5 & 16 \\ -12 & 22 & 15.5 & -1 & 17 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & 0.5 \end{array} \right]$$

Gauss-Jordan method process

- Normalize the pivot equation to make the pivot coefficient to be '1'
- Normalized pivot equation is used to eliminate the off-diagonal terms below and above the pivot equation. The rest is similar to Gauss-elimination.
- Example:

$$\left[\begin{array}{cccc|c} 4 & -2 & -3 & 6 & 12 \\ -6 & 7 & 6.5 & -6 & -6.5 \\ 1 & 7.5 & 6.25 & 5.5 & 16 \\ -12 & 22 & 15.5 & -1 & 17 \end{array} \right]$$

k=1

$$\left[\begin{array}{l} \frac{R_1}{a_{11}} = R'_1 \\ R_2 - (a_{21})R'_1 \rightarrow R'_2 \\ R_3 - (a_{31})R'_1 \rightarrow R'_3 \\ R_4 - (a_{41})R'_1 \rightarrow R'_4 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & -1/2 & -3/4 & 6/4 & 3 \\ 0 & 4 & 2 & 3 & 11.5 \\ 0 & 8 & 7 & 4 & 13 \\ 0 & 16 & 6.5 & 17 & 53 \end{array} \right]$$

Normalize $R_1 \rightarrow R'_1$:

$$\frac{1}{4}(4x_1 - 2x_2 - 3x_3 + 6x_4) = (x_1 - 0.5x_2 - 0.75x_3 + 1.5x_4) = \frac{1}{4}(12) = 3$$

$$a'_{1j} = \frac{a_{1j}}{a_{11}}, \quad b'_1 = \frac{b_1}{a_{11}}, \quad \text{for } j = 1 \text{ to } n$$

Eliminate using R'_1 :

$$R_2: (-6x_1 + 7x_2 + 6.5x_3 - 6x_4) - (-6)(x_1 - 0.5x_2 - 0.75x_3 + 1.5x_4) = (0x_1 + 4x_2 + 2x_3 + 3x_4)$$

$$a'_{2j} = a_{2j} - (a_{21}) a'_{1j} = a_{2j} - (a_{21}) \left(\frac{a_{1j}}{a_{11}} \right), \quad b'_2 = b_2 - (a_{21}) b'_1 \quad \text{for } j = 1 \text{ to } n$$

$$R_3: (x_1 + 7.5x_2 + 6.25x_3 + 5.5x_4) - (1)(x_1 - 0.5x_2 - 0.75x_3 + 1.5x_4) = (0x_1 + 8x_2 + 7x_3 + 4x_4)$$

$$a'_{3j} = a_{3j} - (a_{31}) a'_{1j} = a_{3j} - (a_{31}) \left(\frac{a_{1j}}{a_{11}} \right), \quad b'_3 = b_3 - (a_{31}) b'_1 \quad \text{for } j = 1 \text{ to } n$$

$$R_4: (-12x_1 + 22x_2 + 15.5x_3 - 1x_4) - (-12)(x_1 - 0.5x_2 - 0.75x_3 + 1.5x_4) = (0x_1 + 16x_2 + 6.5x_3 + 17x_4)$$

k=2

$$\left[\begin{array}{cccc|c} 1 & -1/2 & -3/4 & 6/4 & 3 \\ 0 & 4 & 2 & 3 & 11.5 \\ 0 & 8 & 7 & 4 & 13 \\ 0 & 16 & 6.5 & 17 & 53 \end{array} \right] \Rightarrow \left[\begin{array}{l} R'_1 - (a_{12}) R'_2 \rightarrow R''_1 \\ \frac{R'_2}{a_{22}} = R''_2 \\ R'_3 - (a_{32}) R'_2 \rightarrow R''_3 \\ R'_4 - (a_{42}) R'_2 \rightarrow R''_4 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & 0 & -0.5 & 1.875 & 3 \\ 0 & 1 & 2/4 & 3/4 & 11.5/4 \\ 0 & 0 & 3 & -2 & -10 \\ 0 & 0 & -1.5 & 5 & 7 \end{array} \right]$$

Normalize $R'_2 \rightarrow R''_2$:

$$\frac{1}{4}(0x_1 + 4x_2 + 2x_3 + 3x_4) = (0x_1 + x_2 + 0.5x_3 + 0.75x_4) = \frac{1}{4}(11.5)$$

$$a''_{2j} = \frac{a'_{2j}}{a'_{22}}, \quad b''_2 = \frac{b'_2}{a'_{22}}, \quad \text{for } j = 2 \text{ to } n$$

Eliminate using R'_2 : $i=1$ to m , except $i=2$

For $j=2$ to n

$$R'_1 - (-0.5)(0x_1 + 1x_2 + 0.5x_3 + 0.75x_4) = a'_{1j} - (a'_{12}) a''_{2j} = a'_{1j} - (a'_{12}) \left(\frac{a'_{2j}}{a'_{22}} \right)$$

$$R'_3 - (8)(\dots) = a'_{3j} - (a'_{32}) a''_{2j} = a'_{3j} - (a'_{32}) \left(\frac{a'_{2j}}{a'_{22}} \right)$$

$$R'_4 - (16)(\dots) = a'_{4j} - (a'_{42}) a''_{2j} = a'_{4j} - (a'_{42}) \left(\frac{a'_{2j}}{a'_{22}} \right)$$

$$a''_{1j} = a'_{1j} - (a'_{12}) a''_{2j} = a'_{1j} - (a'_{12}) \left(\frac{a'_{2j}}{a'_{22}} \right), \quad b'_2 = b_2 - (a_{21}) b'_1 \quad \text{for } j = 2 \text{ to } n$$

$$a_{1j}^{(k)} = a_{1j}^{(k-1)} - a$$

k=k

Normalize R_k :

*kth row + 1/n
→ 1/5 element / pivot "*

$$\underline{a_{kj}^{(k)}} = \frac{a_{kj}^{(k-1)}}{a_{kk}^{(k-1)}}, \underline{b_k^{(k)}} = \frac{b_k^{(k-1)}}{a_{kk}^{(k-1)}} \quad \text{for } j = k \text{ to } n \quad (\text{When } i \neq k)$$

Eliminate using R_k : For $i=1$ to m , except k

For $j=k$ to n

$$\begin{aligned} R_1^{(k)} : a_{1j}^{(k)} &= a_{1j}^{(k-1)} - a_{1k}^{(k-1)} \cdot a_{kj}^{(k)} = a_{1j}^{(k-1)} - a_{1k}^{(k-1)} \cdot \frac{a_{kj}^{(k-1)}}{a_{kk}^{(k-1)}} \\ R_2^{(k)} : a_{2j}^{(k)} &= a_{2j}^{(k-1)} - a_{2k}^{(k-1)} \cdot \frac{a_{kj}^{(k-1)}}{a_{kk}^{(k-1)}} \\ R_k^{(k)} : a_{kj}^{(k)} &= \frac{a_{kj}^{(k-1)}}{a_{kk}^{(k-1)}} \\ R_m^{(k)} : a_{mj}^{(k)} &= a_{mj}^{(k-1)} - a_{mk}^{(k-1)} \cdot \frac{a_{kj}^{(k-1)}}{a_{kk}^{(k-1)}} \end{aligned}$$

k=m. e.g. m=4

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ R_4^{(4)} = \frac{R_4^{(3)}}{a_{44}} \end{bmatrix} = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & 0.5 \end{array} \right]$$

Exercise:

Write a pseudocode for Gauss-Jordan Elimination

- Augmenting $A(m \times n)$ and $b(m \times 1)$ to become $(m \times n+1)$
-

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for i = 1 to m
  for j = k to n
    
$$a_{ij} = a_{ij} - a_{ik} \cdot \frac{a_{kj}}{a_{kk}}$$

  end j
  
$$b_i = b_i - b_k \cdot \frac{a_{ik}}{a_{kk}}$$

end i
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MATLAB program

See MATLAB demo

LU decomposition

We have learnt Gauss elimination method to find the solution for $\mathbf{Ax} = \mathbf{b}$. We have transformed the matrix $\mathbf{A}$ into an equivalent form of diagonal or upper triangle matrix and transformed the vector $\mathbf{b}$ accordingly in the elimination process.

If we have to solve for different values of $\mathbf{b}$ (i.e., $\mathbf{Ax} = \mathbf{b}_1$, $\mathbf{Ax} = \mathbf{b}_2$, $\mathbf{Ax} = \mathbf{b}_3$, ...) for the same $\mathbf{A}$, then the same Gauss elimination method is repeated for each ' $\mathbf{b}$ ', which is definitely not efficient.

For a more efficient method for finding the solution for multiple $\mathbf{b}$ vectors, we will learn LU decomposition.

- LU decomposition: $\mathbf{A} = \mathbf{LU}$

Using the LU decomposition, we can easily solve for $\mathbf{Ax} = \mathbf{b}$, without finding the inverse of $\mathbf{A}$.

$$\mathbf{Ax} = \mathbf{b} \rightarrow (\mathbf{LU})\mathbf{x} = \mathbf{L}(\mathbf{Ux}) = \mathbf{L}(\mathbf{y}) = \mathbf{b}$$

(Step 1) Solve for $\mathbf{y}$ in $\mathbf{Ly} = \mathbf{b}$, where $\mathbf{y} = \mathbf{Ux}$

- Since $\mathbf{L}$ is lower triangular matrix, $\mathbf{y}$ is easily obtained by forward substitution.

(Step 2) Solve for $\mathbf{x}$ in $\mathbf{Ux} = \mathbf{y}$

- Since $\mathbf{U}$ is upper triangular matrix, $\mathbf{x}$ can be easily solved by backward substitution.

Background on LU decomposition

It is decomposing the matrix $\mathbf{A}$ into $\mathbf{A} = \mathbf{L}\mathbf{U}$

- Upper triangular $\mathbf{U}$
- Lower triangular matrix $\mathbf{L}$

Example:

Initial, $k=0$

$$\mathbf{A} = \mathbf{A}^{(0)} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 4 & 4 & 0 \end{bmatrix}$$

$k=1$

$$\mathbf{A}^{(1)} = \mathbf{M}^{(1)} \mathbf{A}^{(0)}$$

$$\mathbf{A}^{(1)} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & -0.5 & -2 \\ 0 & 2 & -4 \end{bmatrix}$$

$k=2$

$$\mathbf{A}^{(2)} = \mathbf{M}^{(2)} \mathbf{A}^{(1)} = \mathbf{M}^{(2)} \mathbf{M}^{(1)} \mathbf{A}^{(0)}$$

$$\mathbf{A}^{(2)} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & -0.5 & -2 \\ 0 & 0 & -12 \end{bmatrix}$$

$$\mathbf{A} = (\mathbf{M}^{(2)} \mathbf{M}^{(1)})^{-1} \mathbf{A}^{(2)} = \mathbf{L}\mathbf{U}$$

LU decomposition with Gauss Elimination

It is decomposing the matrix $\mathbf{A}$ into $\mathbf{A} = \mathbf{L}\mathbf{U}$

- Upper triangular $\mathbf{U}$
- Lower triangular matrix $\mathbf{L}$

If the upper triangular $\mathbf{U}$ is the output from the gauss elimination, the lower triangular matrix $\mathbf{L}$ describes the process of elimination. You can read other methods such as 'Crout's method' in the textbook.

$$\mathbf{A} \rightarrow \mathbf{U} \quad \text{i.e.} \quad \mathbf{A} = \mathbf{L}\mathbf{U}$$

$$\begin{aligned} k=1, \quad \mathbf{A}^{(1)} &= \mathbf{M}^{(1)} \mathbf{A}^{(0)} &= \mathbf{M}^{(1)} \mathbf{A} \\ k=2, \quad \mathbf{A}^{(2)} &= \mathbf{M}^{(2)} \mathbf{A}^{(1)} &= \mathbf{M}^{(2)} \mathbf{M}^{(1)} \mathbf{A} \\ k=k, \quad \mathbf{A}^{(k)} &= \mathbf{M}^{(k)} \mathbf{A}^{(k-1)} &= \mathbf{M}^{(k)} \mathbf{M}^{(k-1)} \dots \mathbf{M}^{(1)} \mathbf{A} \\ k=m-1 \quad \mathbf{U} &= \mathbf{A}^{(m-1)} = \mathbf{M}^{(m-1)} \mathbf{A}^{(m-2)} &= \mathbf{M}^{(m-1)} \dots \mathbf{M}^{(1)} \mathbf{A} \end{aligned}$$

Therefore, the final row-echelon form $\mathbf{U}$ is expressed in terms of original $\mathbf{A}$ as

$$\begin{aligned} \mathbf{U} &= \mathbf{M}^{(m-1)} \dots \mathbf{M}^{(1)} \mathbf{A} \\ \mathbf{A} &= \left(\mathbf{M}^{(m-1)} \dots \mathbf{M}^{(1)} \right)^{-1} \mathbf{U} \\ &= \left(\mathbf{M}^{(1)} \right)^{-1} \dots \left(\mathbf{M}^{(m-1)} \right)^{-1} \mathbf{U} \\ &= \mathbf{L}\mathbf{U} \end{aligned}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ m_{21} & 1 & 0 & 0 \\ m_{31} & m_{32} & 1 & 0 \\ m_{41} & m_{42} & m_{43} & 1 \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ 0 & 0 & a_{33}^{(2)} & a_{34}^{(2)} \\ 0 & 0 & 0 & a_{44}^{(3)} \end{bmatrix}}_{\text{Output of Gauss elimination}} = \mathbf{L}\mathbf{U}$$

where

$$m_{21} = \frac{a_{21}}{a_{11}}, \quad m_{31} = \frac{a_{31}}{a_{11}}, \quad m_{32} = \frac{a_{32}'}{a_{22}'}, \quad \dots \quad (\text{from Gauss elimination})$$

Derivation

Review Transformation Matrix

Let R_i is the i^{th} row of matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 4 & 1 & 2 \\ 2 & 2 & 4 \\ 8 & 4 & 4 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix}$$

Exercise: What is the output?

$$\begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix}^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix} \qquad \begin{bmatrix} R_2 \\ R_1 \\ R_3 \end{bmatrix}^{(1)} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix}$$

$$\begin{bmatrix} R_1 \\ R_2 - R_1 \\ R_3 \end{bmatrix}^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix}$$

$$\begin{bmatrix} R_1 \\ R_2 - m_{21}R_1 \\ R_3 - m_{31}R_1 \end{bmatrix}^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix}$$

$$= I + \left(\begin{array}{c} k^{th} \text{ row} \\ \text{이제부터 matrix} \end{array} \left(-\frac{a_{ik}}{a_{kk}} \right) \right)$$

Q1) How to get the elements of each $\mathbf{M}^{(k)}$?

Gauss Elimination

K=1

$$\left\{ \begin{array}{l} R_1^{(1)} = R_1^{(0)} \\ R_2^{(1)} = R_2^{(0)} - \frac{a_{21}^{(0)}}{a_{11}^{(0)}} R_1^{(0)} \\ R_3^{(1)} = R_3^{(0)} - \frac{a_{31}^{(0)}}{a_{11}^{(0)}} R_1^{(0)} \\ R_4^{(1)} = R_4^{(0)} - \frac{a_{41}^{(0)}}{a_{11}^{(0)}} R_1^{(0)} \end{array} \right\}$$

K=2

$$\left\{ \begin{array}{l} R_1^{(2)} = R_1^{(1)} \\ R_2^{(2)} = R_2^{(1)} \\ R_3^{(2)} = R_3^{(1)} - \frac{a_{32}^{(1)}}{a_{22}^{(1)}} R_2^{(1)} \\ R_4^{(2)} = R_4^{(1)} - \frac{a_{42}^{(1)}}{a_{22}^{(1)}} R_2^{(1)} \end{array} \right\}$$

K=3

$$\left\{ \begin{array}{l} R_1^{(3)} = R_1^{(2)} \\ R_2^{(3)} = R_2^{(2)} \\ R_3^{(3)} = R_3^{(2)} \\ R_4^{(3)} = R_4^{(2)} - \frac{a_{43}^{(2)}}{a_{33}^{(2)}} R_3^{(2)} \end{array} \right\}$$

LU Decomposition

$$\begin{bmatrix} R_1 \\ R_2 - m_{21}R_1 \\ R_3 - m_{31}R_1 \\ R_4 - m_{41}R_1 \end{bmatrix}^{(1)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -m_{21} & 1 & 0 & 0 \\ -m_{31} & 0 & 1 & 0 \\ -m_{41} & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \end{bmatrix}$$

$$\mathbf{A}^{(1)} = \mathbf{M}^{(1)} \mathbf{A}^{(0)}$$

$$\begin{bmatrix} R_1 \\ R_2 \\ R_3 - m_{32}R_2 \\ R_4 - m_{42}R_2 \end{bmatrix}^{(2)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -m_{32} & 1 & 0 \\ 0 & -m_{42} & 0 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \end{bmatrix}^{(1)}$$

$$\mathbf{A}^{(2)} = \mathbf{M}^{(2)} \mathbf{A}^{(1)}$$

$$= M^2 \cdot (M^1 A^0)$$

$$\begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 - m_{43}R_3 \end{bmatrix}^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -m_{43} & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \end{bmatrix}^{(2)}$$

$$\mathbf{U} = \mathbf{A}^{(3)} = \mathbf{M}^{(3)} \mathbf{A}^{(2)} = M^3 (M^2 M^1 A^0)$$

$$= (M^3 M^2 M^1) A^0$$

Q2 How to solve for $\mathbf{L}$ from $\mathbf{M}^{(k)}$? $\rightarrow$ by multiplying (-1) to m_{ik} elements.

$$\mathbf{A} = (\mathbf{M}^{(m-1)} \dots \mathbf{M}^{(1)})^{-1} \mathbf{U} = \mathbf{L}\mathbf{U}$$

$$\mathbf{L} = (\mathbf{M}^{(m-1)} \dots \mathbf{M}^{(1)})^{-1} = (\mathbf{M}^{(1)})^{-1} \dots (\mathbf{M}^{(m-1)})^{-1}$$

The matrix $\mathbf{M}_k$ from Gauss elimination is a special matrix 'Atomic triangular matrix' that has a special property

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -m_{21} & 1 & 0 & 0 \\ -m_{31} & 0 & 1 & 0 \\ -m_{41} & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{M}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ m_{21} & 1 & 0 & 0 \\ m_{31} & 0 & 1 & 0 \\ m_{41} & 0 & 0 & 1 \end{bmatrix}$$

Therefore, the matrix $\mathbf{L}$ is easily calculated from each elements of $\mathbf{M}^{(k)}$

$$\begin{aligned} \mathbf{L} &= (\mathbf{M}^{(1)})^{-1} \dots (\mathbf{M}^{(m-1)})^{-1} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ -m_{21} & 1 & 0 & 0 \\ -m_{31} & 0 & 1 & 0 \\ -m_{41} & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -m_{32} & 1 & 0 \\ 0 & -m_{42} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -m_{43} & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ m_{21} & 1 & 0 & 0 \\ m_{31} & m_{32} & 1 & 0 \\ m_{41} & m_{42} & m_{43} & 1 \end{bmatrix} \end{aligned} \quad \text{where } m_{ik} = \frac{a_{ik}}{a_{kk}}$$

Exercise: Calculate for the elements of each $\mathbf{M}^{(k)}$ and matrix $\mathbf{U}$, $\mathbf{L}$

$$m_{ik} = \frac{a_{ik}}{a_{kk}} \quad A = \begin{bmatrix} 4 & 1 & 2 \\ 2 & 2 & 4 \\ 8 & 4 & 4 \end{bmatrix}$$

For $k=1$

$$\mathbf{M}^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{a_{21}}{a_{11}} & 1 & 0 \\ -\frac{a_{31}}{a_{11}} & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$k=2, \quad \mathbf{M}^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -m_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{a_{32}}{a_{22}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\mathbf{A} = \mathbf{L}\mathbf{U} = (\mathbf{M}^1)^{-1}(\mathbf{M}^2)^{-1}\mathbf{U} \quad \therefore \mathbf{L} = (\mathbf{M}^1)^{-1}(\mathbf{M}^2)^{-1}$$

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 6.5 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix}$$

Pseudo code for LU decomposition

Exercise:

How would you change the Gauss Elimination Pseudocode to get L, U?

Assume matrix size **A**: $n \times n$, $m=n$

Gauss Elimination	LU Decomposition
<pre>For k=1 to m-1 For i=k+1 to m m(i)=a(i,k) / a(k,k) For j= k to n a(i,j) = a(i,j) - m(i) * a(k,j) End End End</pre>	<pre>Initialize L=eye(n,n) Initialize U=zeros(n,n) For k=1 to m-1 for i=(k+1) to m m(i) = a(i,k) / a(k,k) L(i,k) = m(i) for j=k to n a(i,j) = a(i,j) - m(i)*a(k,j) end j end i end k</pre>

Pseudo code for solving $Lux=b$

- For a square matrix A ($n \times n$), without Permutation

Step 1) LU factorization

Get L & U ^{from matrix A .} using the function
'LU decomposition' above.

$$Ux = b$$

y

Step 2) Solve for y , in $Ly=b$ using Forward Sub.

for $i = 1$ to ~~n~~ n

Sum = 0

for $j = 1$ to $(i-1)$

Sum = Sum + $L(i,j) * y(j)$

end j

$y(i) = b(i) - \text{Sum}$

end i

Step 3) Solve for x , in $Ux=y$ using Backward Sub.

for $i = \del{n}^n$ to 1

Sum = 0

for $j = (i+1)$ to n

Sum += $U(i,j) * x(j)$

end j

$x[i] = (y(i) - \text{Sum}) / U(i,i)$

end i

LU Decomposition with Scale Partial Pivoting ($\mathbf{PA=LU}$)*

For each row elimination process (i.e., for each k), check for the scale partial pivot (k^*) and exchange the rows (row $k \leftrightarrow \text{row } k^*$) of

(1) $\mathbf{A}_k(\mathbf{N}, \mathbf{N})$ (2) Permutation matrix $\mathbf{P}$ (3) $\mathbf{L}$ matrix

Let each row reduction step (k) requires pivoting with permutation matrix $\mathbf{P}_k$

$$\mathbf{A}_1 = \mathbf{A}$$

$$\mathbf{M}_1(\mathbf{P}_1\mathbf{A}_1) = \mathbf{A}_2$$

$$\mathbf{M}_2(\mathbf{P}_2\mathbf{A}_2) = \mathbf{A}_3$$

...

$$\mathbf{M}_k(\mathbf{P}_k\mathbf{A}_k) = \mathbf{A}_{k+1}$$

...

$$\mathbf{M}_{N-1}(\mathbf{P}_{N-1}\mathbf{A}_{N-1}) = \mathbf{A}_N = \mathbf{U}$$

$$\mathbf{A} = (\mathbf{M}_{N-1}\mathbf{P}_{N-1} \dots \mathbf{M}_k\mathbf{P}_k \dots \mathbf{M}_2\mathbf{P}_2\mathbf{M}_1\mathbf{P}_1)^{-1}\mathbf{U}$$

Since $\mathbf{PA=LU}$

$$\mathbf{PA} = \mathbf{P}(\mathbf{M}_{N-1}\mathbf{P}_{N-1} \dots \mathbf{M}_k\mathbf{P}_k \dots \mathbf{M}_2\mathbf{P}_2\mathbf{M}_1\mathbf{P}_1)^{-1}\mathbf{U}$$

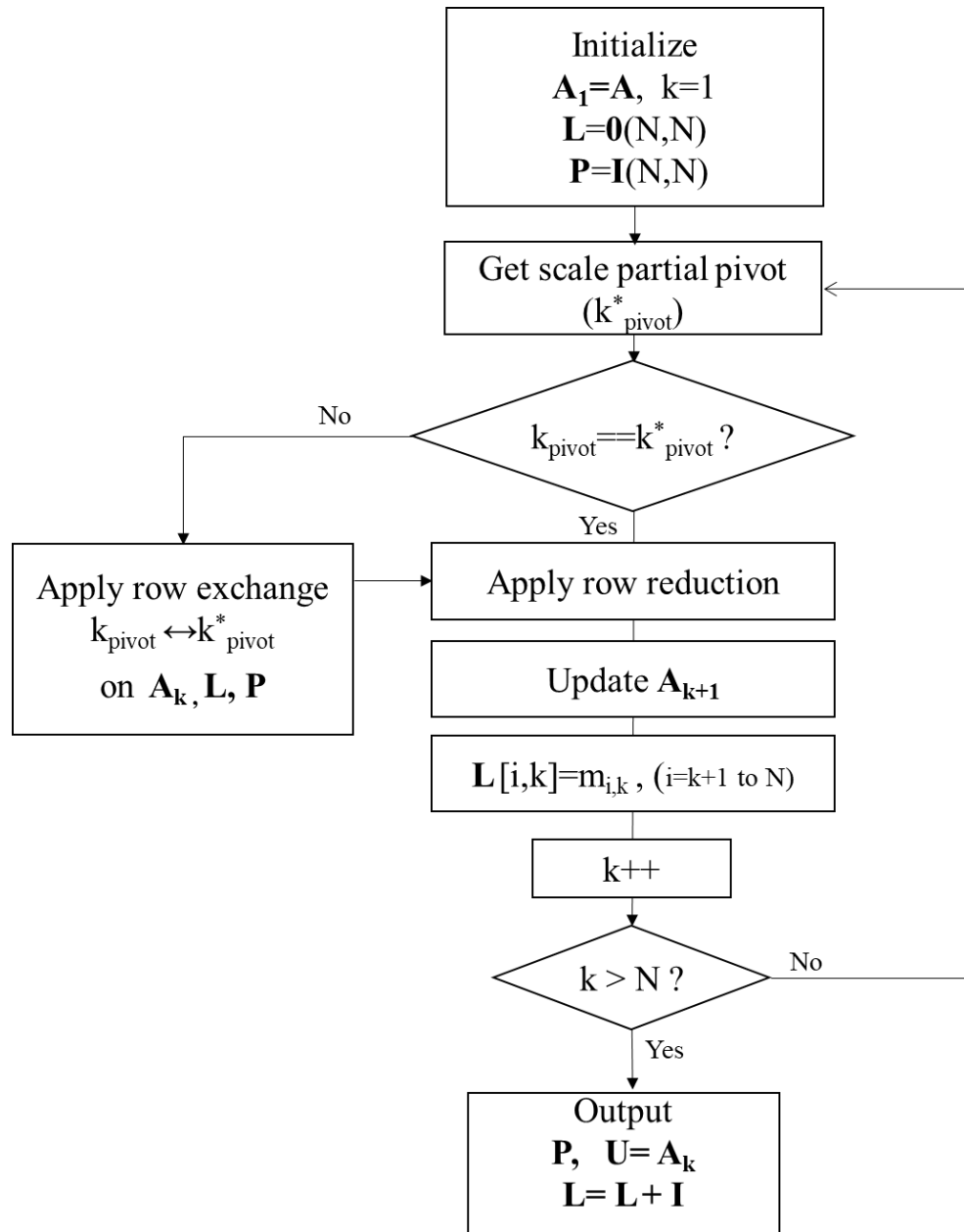
$$\mathbf{L} = \mathbf{P}(\mathbf{M}_{N-1}\mathbf{P}_{N-1} \dots \mathbf{M}_k\mathbf{P}_k \dots \mathbf{M}_2\mathbf{P}_2\mathbf{M}_1\mathbf{P}_1)^{-1}$$

$$\mathbf{L} = \mathbf{P}(\mathbf{P}_1^T\mathbf{M}_1^{-1} \mathbf{P}_2^T\mathbf{M}_2^{-1} \dots \mathbf{P}_k^T\mathbf{M}_k^{-1} \dots \mathbf{P}_{N-1}^T\mathbf{M}_{N-1}^{-1})$$

Flowchart

Assumption: $\mathbf{A}$ is square matrix $\mathbf{A}(N \times N)$.

- Note that matrix $\mathbf{L}$ is initialized as zeros matrix $\mathbf{0}(N, N)$, instead of Identity matrix $\mathbf{I}(N, N)$.



Example

Find LU decomposition of **A** with pivoting in terms of **PA=LU**

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 4 \\ 2 & -0.1 & 3 & -1 \\ -1 & 7 & -4 & 2 \\ 3 & -1 & 4 & 2 \end{bmatrix}$$

Initiation

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

k=1

Pivot $k_{\text{pivot}}^* = 4$. Apply row exchange (row 1 $\leftrightarrow$ row 4)

$$\begin{aligned} \mathbf{P} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \mathbf{P} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \\ \mathbf{A}_1 &= \begin{bmatrix} 1 & 3 & -2 & 4 \\ 2 & -0.1 & 3 & -1 \\ -1 & 7 & -4 & 2 \\ 3 & -1 & 4 & 2 \end{bmatrix} \Rightarrow \mathbf{A}_1 = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 2 & -0.1 & 3 & -1 \\ -1 & 7 & -4 & 2 \\ 1 & 3 & -2 & 4 \end{bmatrix} \end{aligned}$$

After row reduction

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0.667 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.333 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{A}_2 = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 0.567 & 0.333 & -2.333 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 3.333 & -3.333 & 3.333 \end{bmatrix}$$

k=2Pivot $k_{\text{pivot}}^* = 3$. Apply row exchange (row 2 $\leftrightarrow$ row 3)

$$\begin{aligned} \mathbf{P} &= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \mathbf{P} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \\ \mathbf{L} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0.667 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.333 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.667 & 0 & 0 & 0 \\ 0.333 & 0 & 0 & 0 \end{bmatrix} \\ \mathbf{A}_2 &= \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 0.567 & 0.333 & -2.333 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 3.333 & -3.333 & 3.333 \end{bmatrix} \Rightarrow \mathbf{A}_2 = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 0.567 & 0.333 & -2.333 \\ 0 & 3.333 & -3.333 & 3.333 \end{bmatrix} \end{aligned}$$

After row reduction

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.667 & 0.085 & 0 & 0 \\ 0.333 & 0.5 & 0 & 0 \end{bmatrix} \quad \mathbf{A}_3 = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 0 & 0.56 & -2.56 \\ 0 & 0 & -2 & 2 \end{bmatrix}$$

k=3Pivot $k_{\text{pivot}}^* = 4$. Apply row exchange (row 3 $\leftrightarrow$ row 4)

$$\begin{aligned} \mathbf{P} &= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \mathbf{P} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \\ \mathbf{L} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.667 & 0.085 & 0 & 0 \\ 0.333 & 0.5 & 0 & 0 \end{bmatrix} \Rightarrow \mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.333 & 0.5 & 0 & 0 \\ 0.667 & 0.085 & 0 & 0 \end{bmatrix} \\ \mathbf{A}_3 &= \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 0 & 0.56 & -2.56 \\ 0 & 0 & -2 & 2 \end{bmatrix} \Rightarrow \mathbf{A}_3 = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0.56 & -2.56 \end{bmatrix} \end{aligned}$$

After row reduction

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0 & 0 \\ 0.333 & 0.5 & 0 & 0 \\ 0.667 & 0.085 & -0.28 & 0 \end{bmatrix} \quad \mathbf{A}_4 = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$k=(4==N)$$

$$U = A_4, \quad L = L + I$$

Output for PA=LU

$$A = \begin{bmatrix} 1 & 3 & -2 & 4 \\ 2 & -0.1 & 3 & -1 \\ -1 & 7 & -4 & 2 \\ 3 & -1 & 4 & 2 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -0.333 & 1 & 0 & 0 \\ 0.333 & 0.5 & 1 & 0 \\ 0.667 & 0.085 & -0.28 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 3 & -1 & 4 & 2 \\ 0 & 6.667 & -2.667 & 2.667 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

Q: How would the solution $\mathbf{x}$ change if we had to do pivoting in gauss elimination in the process of LU decomposition? Use the permutation matrix $\mathbf{P}$.

- Permutation matrix $\mathbf{P}$ is needed to record the exchanges of row equations
 $A=LU$. If pivoting was carried out
 $\Rightarrow PA=LU, \quad PAx=Pb$
- $PA=LU$
- $Ax=b \rightarrow PAx=Pb \rightarrow PAx=d$
- L, U are output with pivoting applied.
- Use $\mathbf{P}^*\mathbf{b}$ instead of $\mathbf{b}$ in solving for $\mathbf{x}$

: solve for $\mathbf{y}$ in $\mathbf{Ly}=\mathbf{P}^*\mathbf{b}$, where $\mathbf{y}=\mathbf{Ux}$. Then, solve for $\mathbf{x}$ in $\mathbf{Ux}=\mathbf{y}$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ z \\ x \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ x \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ x \\ z \end{bmatrix}$$

Pseudo code for LU decomposition with Permutation

Exercise:

- For a square matrix A ($n \times n$), with Permutation

Step 1) LU factorization

$A(n \times n)$ is given

$A \rightarrow L, U, P$

$$PAx = Pb = LUx = b'$$

$$PA = LU$$

$$L(Ux) = Ly = Pb = b'$$

y

$$Ly = Pb = b'$$

Step 2) Solve for y , in $Ly = b'$ using Forward Sub.

for $i = 1$ to n

sum = 0

for $j = 1$ to $(i-1)$

sum = sum + $L(i, j) * y(j)$

end j

$y(i) = b'(i) - \text{sum}$

end i

Step 3) Solve for x , in $Ux = y$ using Backward Sub.

for $i = n$ to 1

sum = 0

for $j = (i+1)$ to n

sum += $U(i, j) * x(j)$

end j

$x[i] = (y(i) - \text{sum}) / U(i, i)$

end i

Write a C-program

See assignment instructions

```
void LLdecomp ( Matrix A, Matrix L, Matrix U, Matrix P)
```

```
// step 1 : LL factorization
```

$$Ax = b \rightarrow PAx = P\underline{LU}x = Pb \quad \therefore Pb = b'$$

```
// step 2 : Ly = b fwd substitution
```

```
for(int i=1; i <= n; i++) {
```

```
    sum = 0;
```

```
    for(int j=1; j < i; j++) {
```

```
        sum += L.at[i][j] * y.at[j][0];
```

```
    }
```

```
    y.at[i][0] = b'.at[i][0] - sum;
```

```
}
```

```
// Step 3 : Ux = b backward substitution
```

```
for(int i=n; i >= 1; i--) {
```

```
    sum = 0;
```

```
    for(int j=(i+1); j <= n; j++) {
```

```
        sum += U.at[i][j] * x[j][0];
```

```
    }
```

```
    x.at[i][0] = ( y.at[i][0] - sum ) / U.at[i][i];
```

```
}
```

Inverse Matrix

We have solved $\mathbf{Ax}=\mathbf{b}$, with Gauss Elimination without finding the matrix inverse.

Another approach of solving the linear system is using the inverse of matrix $\mathbf{A}$, if it is invertible.

There are several methods for finding the matrix inverse but this lecture will cover (1) LU decomposition and (2) Gauss-Jordan method

$\mathbf{A}^{-1}$ from LU decomposition

$$\text{Let } \mathbf{A}^{-1} = \begin{bmatrix} x_{11} & \cdots & \cdots & x_{1n} \\ x_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ x_{n1} & \cdots & \cdots & x_{nn} \end{bmatrix}$$

Using the fact that $\mathbf{AA}^{-1} = \mathbf{I}$,

$$\mathbf{A} \begin{bmatrix} x_{11} \\ \vdots \\ \vdots \\ x_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{A} \begin{bmatrix} x_{12} \\ \vdots \\ \vdots \\ x_{n2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots \quad \mathbf{A} \begin{bmatrix} x_{1n} \\ \vdots \\ \vdots \\ x_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Solve n systems with **LU** decomposition to find $\mathbf{A}^{-1}$

Note that $\mathbf{A} = \mathbf{LU}$ is calculated only once, and only the **forward substitution** and **backward substitution** are implemented n times.

$\mathbf{A}^{-1}$ from Gauss-Jordan method

- Augment $\mathbf{A}$ with $\mathbf{I}$ as $[\mathbf{A} \mid \mathbf{I}]$
- Then, process with the Gauss-Jordan method and the result will be

$$[\mathbf{I} \mid \mathbf{A}^{-1}]$$